

Harshith Goka

Research Engineer, Samsung Research Korea

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Experience

Samsung Research

AI System Optimization

AI Model Taskforce

- On Device LLM Taskforce
- Language Intelligence Team
- Visual Intelligence Team

Samsung Research

- Internship @ Vision Team

Seoul, South Korea

February 2025 - Current

October 2025 - January 2026

February 2024 - January 2025

March 2023 - February 2024

September 2019 - March 2023

Seoul, South Korea

Summer 2018

Education

Indian Institute of Technology, Bombay

- B.Tech. in Computer Science and Engineering; **CGPA: 8.74/10**

Mumbai, India

July 2015 - June 2019

- Major: **Computer Science and Engineering**
- **Key Electives:** Foundations of Intelligent and Learning Agents, Advanced Machine Learning, Foundations of Machine Learning, Computer Vision, Design and Re-engineering of Object Oriented Programming, Parallel Programming Paradigms, Foundations of Digital Image Processing
- **Core Courses:** Artificial Intelligence, Implementation of Programming Languages, Automata Theory, Discrete Structures, Data Analysis and Interpretation, Abstractions and Paradigms for Programming, Databases and Information Systems, Software Systems Lab, Operating Systems, Data Structures and Algorithms, Design and Analysis of Algorithms, Computer Networks, Digital Logic Design, Computer Architecture

Skills

- **Languages:** Python, C/C++, Java, Kotlin, L^AT_EX, Dart, JavaScript, SQL, VHDL, C#
- **Frameworks:** Android SDK, MATLAB, Docker, Flutter, Qualcomm AI Direct Engine(QNN)
- Experienced in using popular deep learning libraries **PyTorch**, **JAX**, **Tensorflow**, **Transformers**, **Megatron** on multinode HPC clusters

Key Projects

Next On-Device LLM Model Development

Samsung Research, Korea

- Leader: Tang Aiden/Lee Kyenghun

Oct 2025 - Jan 2025

- Implemented nested sub-model architectures to enable elastic inference, allowing for task-dependent model scaling during deployment
- Engineered high-fidelity nested sub-models for on-device environments, balancing training-time penalties with preserved full-scale performance and inference elasticity

- On-Device LLM Neural Architecture Search** Samsung Research, Korea
June 2025 - Sept 2025
 - Leader: Lee Kyenghun
 - Developing a framework for automatic configuration of architectures for pretraining LLM models using NeMo.
 - Conducting various heuristic based model structure trainings for improving the quality while reducing latency or model size.
 - Also using various genetic search algorithms to automatically improve upon existing architectures for on device LLM architectures which are both fast and high quality.
- Optimization of On-Device Architectures** Samsung Research, Korea
Feb 2025 - Current
 - Leader: Lee Kyenghun
 - Developed an automated framework for NPU latency profiling.
 - Optimized SSM, hybrid, and prototype architectures for efficient on-device inference.
 - Accelerated speech encoder and TTS diffusion models for NPU deployment.
- Quantization for On Device LLMs** Samsung Research, Korea
March 2024 - July 2024
 - Leader: Lee Kyenghun
 - Worked on quantization of LoRA with Samsung Gauss On-Device models which made to production on Samsung Galaxy devices
 - Improved various tools used for quantization of deep models.
- Object Eraser - Video Inpainting** Samsung Research, Korea
February 2022 - December 2022
 - Leader: Naejin Kong
 - Explored video inpainting models using Transformer in the Generative Adversarial Network (**GAN**) framework.
 - Experimented on various **Vision Transformers** model types to improve the quality.
- Object Eraser - Image Inpainting** Samsung Research, Korea
March 2021 - December 2021
 - Leader: Naejin Kong
 - Explored various ways to use Generative Adversarial Networks (**GANs**) to inpaint part of image to match the background.
 - Co-developed a new way of using Fast Fourier Convolutions to improve the quality of large mask inpainting.
 - Developed various sized on-device models to run on mobile phones and laptops using **TFLite** and **ONNXRuntime**.
 - Commercialized the Object Eraser feature on the Gallery app on Samsung Galaxy Book devices.
- Image Relighting** Samsung Research, Korea
September 2019 - March 2020
 - Leader: Naejin Kong
 - Explored various ways of relighting an image by detecting types of materials and re-rendering them.
 - Generated a large amount of synthetic data for detecting materials.
 - Trained deep U-net models to extract various component maps like albedo, normal and roughness maps.
- Attention Methods for Deep Metric Learning** Samsung Research, Korea
Summer 2018
 - Guide: Wonsik Kim
 - Explored different soft attention methods for deep metric learning on images.
 - Developed GPU code for a few attentions models using **Caffe**.
 - Evaluated these different models on the dataset of clothes using the **Recall@K** metric.